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Aspects of pre-Tertiary paleomagnetism of Spitsbergen and their tectonic implications *

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Abstract

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Paleomagnetic investigations were carried out on Cretaceous dolerite dykes and sills, Permian limestones, Permo-Carboniferous limestones and Lower Devonian sandstones from different parts of Spitsbergen. Detailed rock magnetic studies showed that the main magnetic minerals in dolerites, limestones and sandstones are titanomagnetite, magnetite and hematite respectively. The rocks were demagnetized thermally and in alternating fields. Vector analysis of the demagnetization path separates the different components of NRM, and establishes a characteristic remanence. The paleomagnetic poles for Cretaceous dolerites, Permo-Carboniferous limestones and Lower Devonian sandstones from Spitsbergen, are respectively: $\lambda' = 65°\,N$, $\phi' = 222°\,E$, $\lambda' = 34°\,N$, $\phi' = 140°\,E$ and $\lambda' = 12°\,N$, $\phi' = 135°\,E$. When compared with the European and North American apparent polar wander paths, all Spitsbergen poles lie at a distance from both APWPs suggesting that Spitsbergen has been rotated and even translated in relation to Europe during the opening of the Labrador Sea and Greenland–Norwegian Sea.

Introduction

Rocks exposed in Spitsbergen, the main island of the Svalbard Archipelago, show a dominant N–S oriented structural alignment which reflect the superposition of four main episodes of deformation. The most intensive episode was the Caledonian orogeny (Harland, 1969; Birkenmajer, 1981). The three later tectonic events were less violent and have not involved any metamorphism. In the late Devonian, the Svalbardian tectonic stage affected the area causing N–S trending fracturization, faulting and some folding. The Variscan orogeny was insignificant and largely rejuvenated older structural trends. In the Tertiary, folding and thrusting of the western margin of Spitsbergen were connected with compression resulting from the dextral movement of Greenland during the opening of the Greenland–Norwegian Sea. In this context, Spitsbergen with its long geological history offers a potential for correlation between mobile and stable Europe and for tracing plate tectonic events.

The objective of this paper is to summarize the paleomagnetic investigations carried out by the author in Spitsbergen during the last few years from this point of view.

Sampling areas and experimental procedure

The paper summarizes paleomagnetic investigations and rock magnetic studies of Mesozoic dolerite dykes and sills from west and east Spitsbergen (Jeleńska et al., 1978/1979; Vincenz et al., 1981; Vincenz et al., 1984), Permo-Carboniferous limestones from west Spitsbergen (Jeleńska, 1985;

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Vincenz and Jeleńska, 1985) and Lower Devonian red sandstones from central Spitsbergen (Løvlie et al., 1984; Jeleńska and Lewandowski, 1986). Preliminary results for Permian limestones from central Spitsbergen are also included.

As mentioned above, during the Tertiary Spitsbergen was subject to strong tectonic deformation which affected the western margin of the post-Caledonian platform and formed the 10–20 km wide West Spitsbergen Orogenic Front. An attempt was made to collect samples from locations both within and outside this Tertiary orogenic belt (Fig. 1b).

Mesozoic dolerite dykes were sampled within the Tertiary Orogenic Front at Hornsund, southwest Spitsbergen, whereas sills were sampled from outside the Front on the east coast. Detailed rock magnetic studies of dolerite dykes and sills by Jeleńska et al., (1978/1979) comprise thermomagnetic analyses, acquisition of TRM and IRM for determination of domain state of magnetic minerals (Lowrie and Fuller, 1971), the Hopkinson effect and microscopic analysis of polished sections in reflected light. It has been stated (Jeleńska et al., 1978/1979; Vincenz et al., 1981; Vincenz et al., 1984) that dolerite dykes and sills contained medium-to-fine grained titanomagnetite, relatively little altered by deuteric oxidation, with some overprint of hydrothermal changes. The magnetic minerals show a bimodal distribution of grain size with a predominance of single domain (SD) grains which are the carriers of NRM of thermomagnetic origin. Figure 2a shows an example of thermomagnetic analysis based on the decay of saturation IRM during heating in the field-free space.

The sample was magnetized in a field of 1.6 T and the decay of saturation IRM was recorded during heating in a compensated field. After cooling to room temperature the sample was remagnetized by the same field and the decay curve was obtained during a second heating.

The directional changes of NRM during thermal and A.F. demagnetization together with Zijderveld diagrams are the basis of an estimation of the final directions of remanence. Examples of Zijderveld diagrams and demagnetization curves are shown in Fig. 3a. The validity of the selected

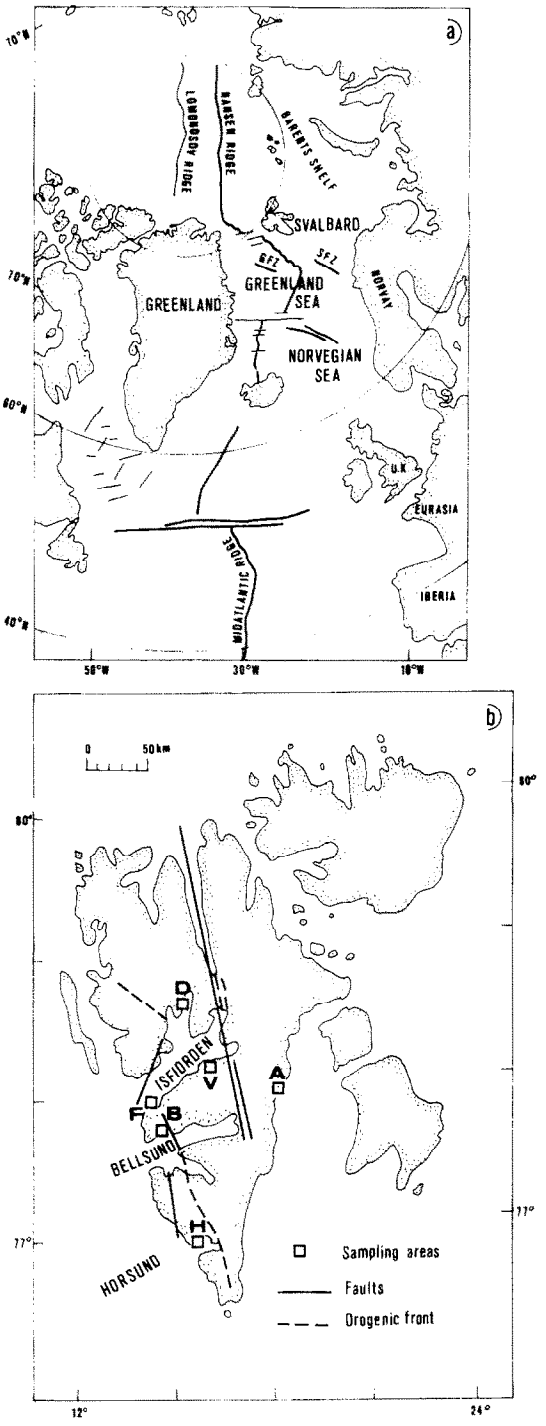
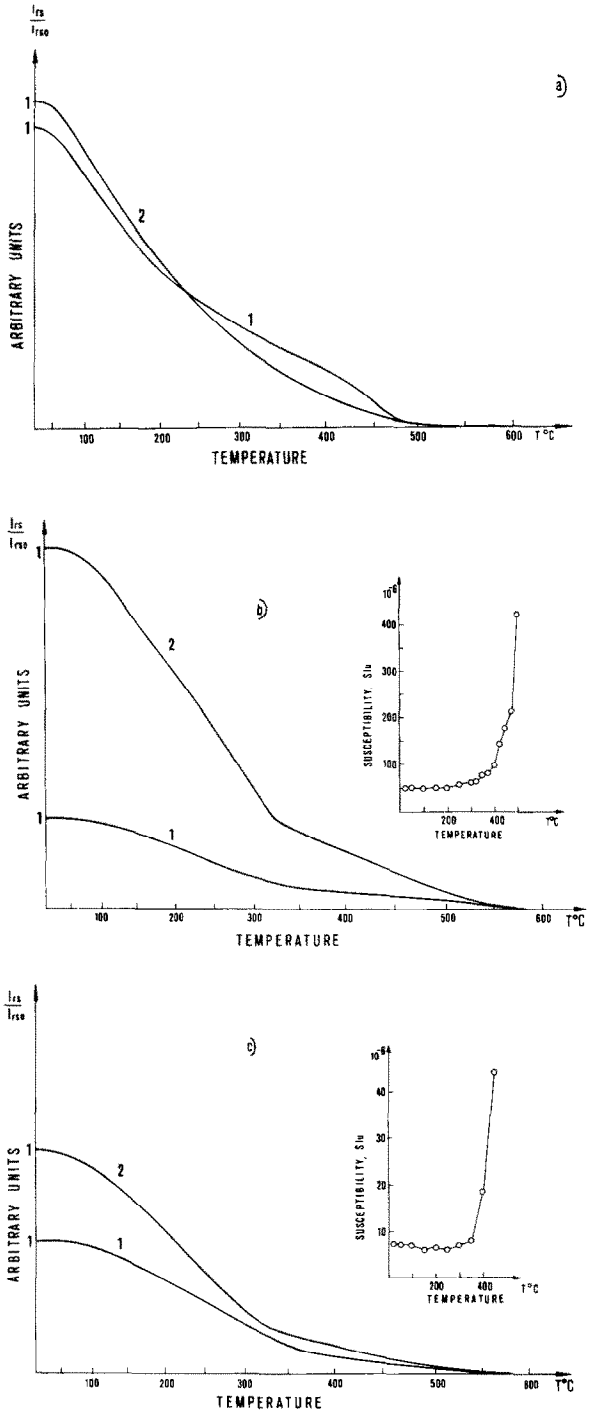


Fig. 1. Map of Spitsbergen, Svalbard Archipelago. a. Location of Svalbard in the Arctic region. b. Sampling areas: *A*—Agardhbukta, Mesozoic dolerite sills: *H*—Hornsund, Mesozoic dolerite dykes; *V*—Vindodden, Permian limestones; *B*—Bellsund, Permo-Carboniferous limestones; *F*—Festningen, Permo-Carboniferous limestones; *D*—Dicksonfjorden, Devonian sandstones. *SFZ*—Spitsbergen Fracture Zone, *GFZ*—Greenland Fracture Zone.

directions has been checked by a least squares method of analysis of step demagnetization data (Stupavsky and Symons, 1978). The method is based on the assumption that the intensity of each component decays exponentially during demagnetization. The mean directions for dykes and sills



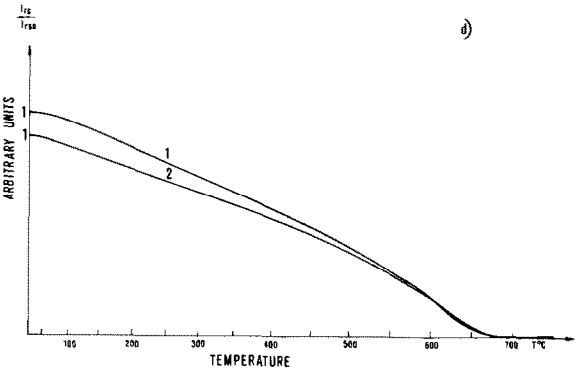


Fig. 2. Examples of decay curves of saturation remanence during heating. a. Mesozoic dolerites. b. Permo-Carboniferous limestones. c. Permian limestones. d. Devonian sandstones. *1* —first heating; *2*—second heating. In Figs. b and c magnetic susceptibility after subsequent heating is also shown.

are shown in Fig. 4. The paleomagnetic poles for dykes and sills are $\lambda' = 70°\,\mathrm{N}$, $\phi' = 223°\,\mathrm{E}$ and $\lambda' = 54°\,\mathrm{N}$, $\phi' = 225°\,\mathrm{E}$, respectively.

As magnetic mineralogy, environmental conditions of deposition and behavior of remanence during demagnetization are the same for both Permo-Carboniferous limestones from the west coast of Spitsbergen and Permian limestones from central Spitsbergen, both rock collections have been described together. According to Siedlecka (1970) the Permian and uppermost Carboniferous sediments consist mainly of marly, arenaceous and cherty limestones. The study of magnetic properties was limited to only a few experiments because

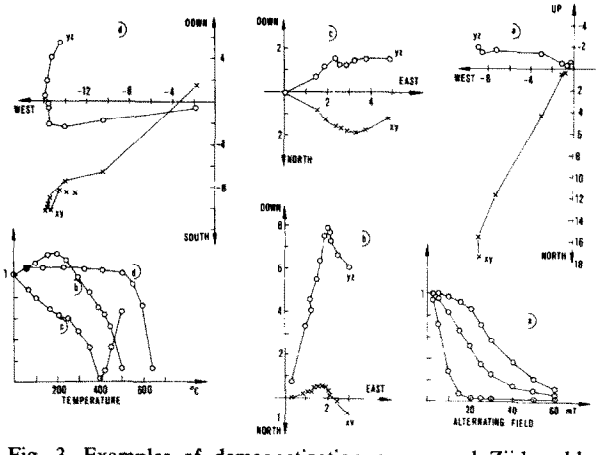


Fig. 3. Examples of demagnetization curves and Zijderveld diagrams. a. Mesozoic dolerites. b. Permo-Carboniferous limestones. c. Permian limestones. d. Devonian sandstones. Units on vector axis 8×10^{-5} A m^{-1}. Coordinates after tilt correction.

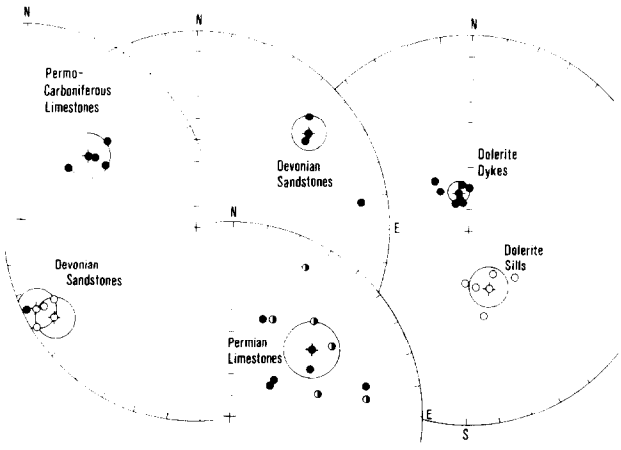


Fig. 4. Mean directions for Mesozoic dolerites, averaged for dykes and sills, Permo-Carboniferous and Permian limestones, averaged for lithostratigraphic formations and sites, respectively, and Devonian sandstones, averaged for localities. Crossed circles—mean directions for a rock unit given with a circle of 95% confidence. For Devonian sandstones mean of normal and reversed directions are shown in addition to mean of all directions; full circles—normal polarity; open circles—reversed polarity; half-open circles—mixed polarity. The directions are after tilt correction.

of the low concentration of magnetic material. A large amount of pyrite was found, revealed in polished sections when examined under a microscope in reflected light. The presence of magnetite, probably the main carrier of NRM, was observed during the heating of saturation IRM in a field-free space (Fig. 2b, c) and during thermal demagnetization of NRM (Fig. 3b, c). The saturation IRM of unheated samples (first curves) was so weak that it was difficult to exactly determine the value of the Curie temperature and to distinguish between magnetite and maghemite. After first heating to 600 °C IRM increased between a hundred to a thousand times and the Curie temperature of newly created minerals could be determined (Vincenz and Jeleńska, 1985; Jeleńska, 1985). Maghemite shows a decrease in susceptibility after heating above 300 °C due to conversion to hematite (Lowrie and Heller, 1982). Such a decrease (Fig. 2b, c) or hematite is not observed, excluding the presence of maghemite. The observed magnetite might be of primary origin or derived from pyrite. The latter possibility has been considered because of the observation that during the heating of rock up to 600 °C new magnetite was created, very often together with pyrrhotite. Both might be a product of the conversion of pyrite which was present in the unheated rock. According to Ward (1970) and Schwarz (1975) slow oxidation of hexagonal pyrrhotites leads to the formation of monoclinic magnetic pyrrhotite and magnetite. In natural rocks containing pyrite it is often observed that pyrite converts to magnetite and magnetic pyrrhotite (Tsusue, 1962; Turner, 1975; Lowrie and Heller, 1982).

Thermal and A.F. demagnetization of the NRM showed complex and multicomponent behaviour. Examples of Zijderveld diagrams and demagnetization curves are demonstrated in Fig. 3b and c. The step demagnetization data were analysed by fitting lines and planes to the demagnetization path of a specimen. The calculations were made by means of Kirschvink's program (1980). Two groups of directions were separated in the Permian and Permo-Carboniferous limestones: (1) steep directions of secondary origin, close to the Mesozoic paleofield found in the dolerite dykes and sills; (2) shallow, down-dipping remanence, which might be the Permian direction in Spitsbergen. The directions for Permian and Permo-Carboniferous limestones are shown in Fig. 4. The paleomagnetic pole for Permo-Carboniferous limestones from the Tertiary Orogenic Front on the west coast of Spitsbergen agrees well with the paleomagnetic pole for the Permian limestones from the tectonically undisturbed region of central Spitsbergen, being respectively: $\lambda' = 36°\,\mathrm{N}$, $\phi' = 141°\,\mathrm{E}$ and $\lambda' = 31°\,\mathrm{N}$, $\phi' = 140°\,\mathrm{E}$.

Devonian sandstones of the Wood Bay Formation (Lower–Middle Devonian) were sampled in Dicksonfjorden, central Spitsbergen, not within the West Spitsbergen Orogenic Front. Rock magnetic properties such as IRM acquisition, thermomagnetic analysis, magnetic susceptibility and rotational hysteresis measurements indicate a fine-grained hematite (Fig. 2d). Petrographic observations suggest a detrital origin for the hematite granules. These observations, together with almost coincidental azimuthal distribution of remanent directions and axis of maximum susceptibility, suggest a depositional origin of magnetization carried by hematite grains.

Examination of the demagnetization curves,

TABLE 1

The main results of paleomagnetic and rock magnetic investigations *

Age	Area and rock type	Magnetic carriers	Demagnetization technique	N/n	Dec. (°)	Inc. (°)	k	α_{95} (°)	ϕ' (°)E	λ' (°)N
Cretaceous 110 m.y.	Hornsund dolerite dykes	SD grains of titanomagnetite, slightly altered hydrothermally	A.F. thermal	7/55	343	72	193	5	223	70
Cretaceous 100 m.y.	Agardhbukta dolerite sills	SD grains of titanomagnetite, slightly altered hydrothermally	A.F. thermal	5/42	159	−62	56	9	225	54
Permian	Vindodden limestones	fine-grained magnetite of detrital origin or derived from pyrite	A.F. thermal [a]	2/11	50	42	12	14	140	31
Permo-Carboniferous	Festningen Bellsund limestones	fine-grained magnetite of detrital origin or derived from pyrite	A.F. thermal [a]	4/35	48	46	85	10	141	35
Lower Devonian	Dicksonfjorden red beds sandstones	fine-grained hematite of detrital origin	A.F. thermal [a]	6/15	238	−12	44	10	135	12

* N—number of localities, rock units or formations, n—number of sites. The directions are after tilt correction.

[a] pilot specimens only.

Zijderveld diagrams and directional changes (Fig. 3d) similarly revealed characteristic directions which were checked by calculation of lines and planes fitted to the demagnetization path. The mean directions for the Devonian sandstones are shown in Fig. 4. The paleomagnetic pole for the Devonian sandstones is $\lambda' = 12°\,\mathrm{N}$, $\phi' = 135°\,\mathrm{E}$. Table 1 summarizes briefly the main results.

Discussion

The paleomagnetic poles obtained for Mesozoic, Permo-Carboniferous and Devonian rocks from Spitsbergen were compared with the apparent polar wander path for Europe and North America (Fig. 5). The curves for Cretaceous to Permian samples were calculated by Westphal et al. (1986), whereas the Carboniferous, Devonian and Silurian poles were calculated by the author on the basis of the catalogues of paleomagnetic poles (Hicken et al., 1972; Irving et al., 1976; McElhinny and Cowley, 1977; 1980; Khramov, 1982) and from more recent publications (Abrahamsen, 1977; Thorning and Abrahamsen, 1980; Torsvik et al., 1983; Torsvik, 1984). Up to the Upper–Middle Devonian, poles were averaged for both the British Isles and continental Europe. Lower Devonian and Upper Silurian poles were given only for the British Isles because there is no data from continental Europe for that period. The only Silurian pole from Norway (Irving et al., 1976) is given as an alternative for the continuation of European APWP.

As shown in Fig. 5, all Spitsbergen poles lie far from the European and North American APWPs, except the Lower–Middle Devonian pole which approaches the British one. This difference between Spitsbergen and European poles is displayed in Fig. 6. The differences in paleolatitude are more or less the same for all Paleozoic data,

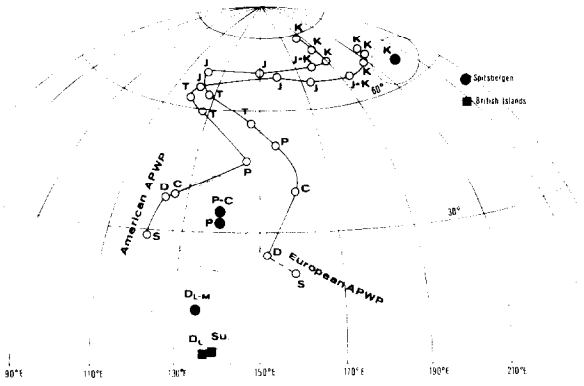


Fig. 5. Comparison of Spitsbergen paleomagnetic poles with the European and American APWPs. The Spitsbergen poles are shown with a circle of 95% confidence.

whereas the deviation of declination is in the opposite direction for the Mesozoic. This indicates two movements: the first one taking place somewhere during the Permian and Cretaceous and the second one after the Cretaceous. These movements can be associated in time with the following events in the Arctic region: (1) the beginning of the opening of the North Atlantic, when Greenland was detached from the North American plate, and (2) the creation of the Greenland–Norwegian Sea and the separation of Spitsbergen from Greenland.

The movement of Spitsbergen relative to the European plate is clearly evident from the paleomagnetic results, but is found difficult to accept by geologists. There is no geological evidence for a displacement between Europe and Spitsbergen since the Late Devonian while, according to Harland (1978), there were three pre-Devonian provinces in Svalbard, widely separated along a N–S line, which were brought together by strike-slip motion along the Balliolbreen Fault. There is also geophysical evidence against the mobility of Spitsbergen relative to Europe. The Barents Sea area between Spitsbergen and Norway, according to Davydova et al. (1985), is a deep sedimentary basin similar in the thickness of the sedimentary layer to those of the East European platform, based on a deep seismic profile, which varies from 8 to 17 km. The numerous earthquakes recorded in recent years in northern and eastern Spitsbergen (Bungum et al., 1982; Chan and Mitchell, 1985) are interpreted by Chan and Mitchell (1985) as intraplate activity occurring along existing faults. On the other hand, support for the mobility of Spitsbergen can also be found. Savostin and Karasik (1981) associated the same Spitsbergen earthquakes with a plate margin which separates the Spitsbergen microplate from the Eurasian plate. The paleogeographic reconstruction of Greenland and Europe relative to North America made from magnetic anomalies (Vink, 1982; Srivastava, 1985) shows a large overlap between Spitsbergen and northeast Greenland. This is explained by Vink (1982) as a result of a rifting process which led to continental extension and thinning of the crust. Seismic refraction measurements from the Kara Sea and Yermak Plateau, when compared with the results from Barents Sea (Mair and Forsyth, 1982), show a thinning of the crust in the east and southeast of Spitsbergen. However, the overlap between Spitsbergen and Greenland can be explained by a rotation of Svalbard relative to Europe. In that case, the Bjørnøyrenna and Hammerfest basins, lying in the southwest part of the Barents shelf, can be regarded as a failed arm of a triple junction (Birkenmajer, 1981). The remaining arms, the De Geer and Harland Lines, have been active.

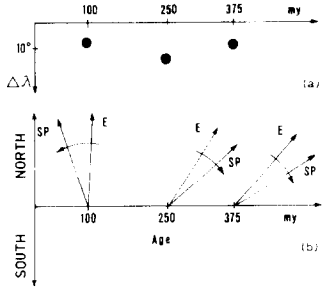


Fig. 6. Difference in paleolatitude (a) and declination (b) between Spitbergen and Europe for Mesozoic dolerites (100 m.y), Permo-Carboniferous limestones (250 m.y) and Devonian sandstones (375 m.y). Δλ — difference of paleolatitude; *Sp* — Spitsbergen; *E* — European plate. Declination in clockwise direction from the north.

Summing up one may state that the paleomagnetic results suggest the mobility of Spitsbergen relative to Europe. The alternative explanation that all Spitsbergen rocks have been partly remagnetized during the West Spitsbergen Orog-

eny during the Tertiary, although less possible, cannot be excluded.

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